

Four Months' Notice

Turning El Niño into a dengue early-warning alarm for ASEAN

A freely available climate index gives ASEAN health offices months of warning before a dengue season turns severe. We turn that lag into a two-condition alert rule any provincial health office can run today — no model, no code, no paid data.

SDG 3 Health

SDG 13 Climate

SDG 6 Water

Team Swallowgank | Institut Teknologi Sepuluh Nopember | Indonesia

THE PROBLEM

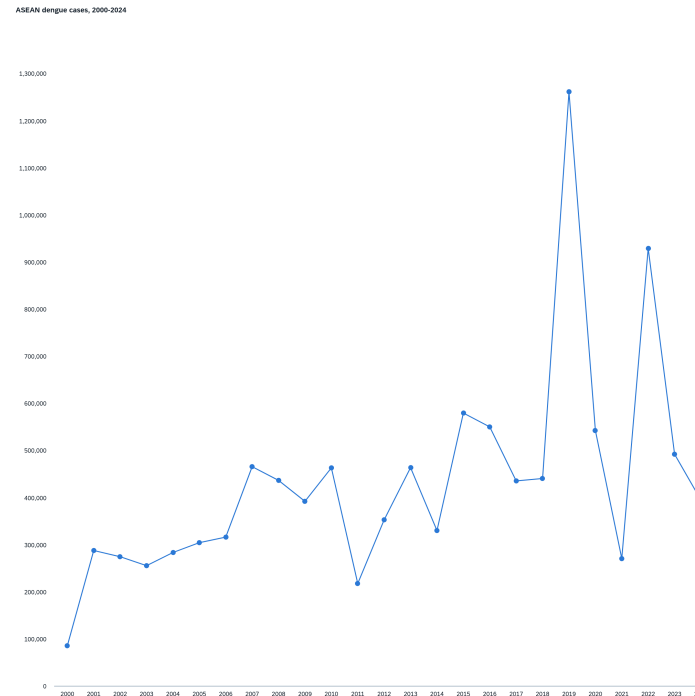
Dengue is ASEAN's fastest-moving climate-sensitive disease

1.26M

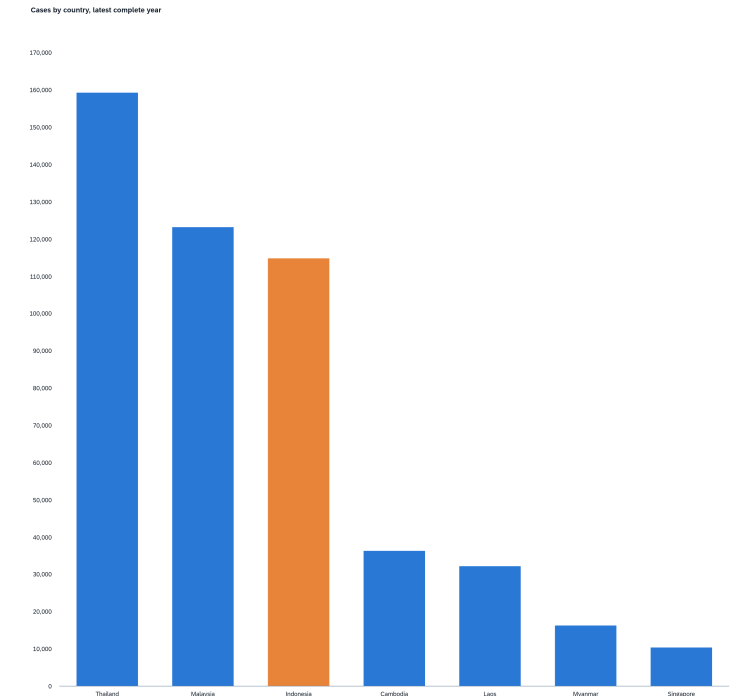
cases in 2019 — the worst year in our nine-country panel

- Vietnam 368k · Philippines 220k · Thailand 159k (latest complete year per country)
- The 2023–24 El Niño added an estimated **9.6 million cases worldwide** (Tian et al., 2025)
- Outbreaks are still fought **reactively** — after cases surge

ASEAN dengue cases, 2000–2024



Cases by country, latest complete year



THE QUESTION

Climate data arrives months before patients do

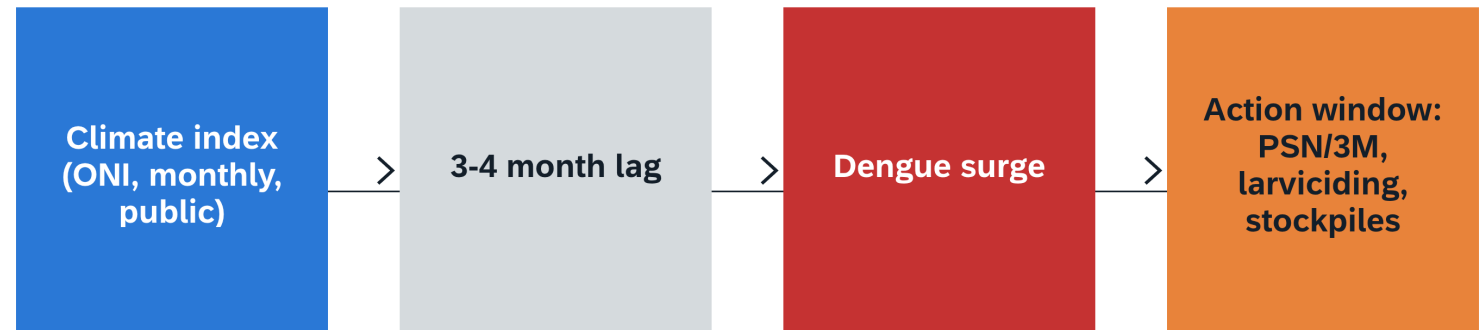
Can a freely available climate index tell a health office how bad the coming season will be — early enough to act?

- **SDG target 3.3** — end the epidemics of neglected tropical diseases; dengue is one of them
- **SDG target 13.1** — resilience and adaptive capacity to climate-related hazards
- **SDG target 6.1** — safe water for all; without piped water households store it, and *Aedes* breeds in the store

ASEAN priority. Dengue sits under **ASEAN Health Cluster 2** — *Responding to All Hazards and Emerging Threats* — whose ASEAN Dengue Day statements ask member states for earlier, multi-sector action.

Our test: does the signal survive when we remove the seasonal cycle, hold out unseen years, and use the WHO outbreak definition?

From public index to public action



DATA & METHOD

Nine open datasets, one monthly panel, zero paid sources

DATA	ROWS	COVERAGE
Dengue cases (OpenDengue V1.3)	70,397	10 countries, 1955–2025
Dengue × climate monthly panel	55,329	province-month, lags pre-computed
Temperature & rainfall (CCKP, ERA5)	9,000	monthly, 1950–2024
ONI El Niño index (NOAA CPC)	918	monthly, 1950–2026
Water & sanitation (WHO/UNICEF JMP)	8,692	2000–2024
Population (BPS Indonesia)	140	provinces, denominators
Flood exposure (GHSL)	30,348	836 cities, 1975–2030

SDG 3

SDG 13

SDG 6

Every table joins on the same two keys — iso3 + year (plus period for the monthly dengue × climate panel). Nothing is entered by hand.

METHOD

- Monthly national series, 2010+
- Lagged correlations, 0–4 months
- **Anomalies** — seasonal cycle removed before we believe any correlation
- A two-condition trigger rule
- Out-of-sample test on unseen years
- WHO/PAHO endemic-channel benchmark

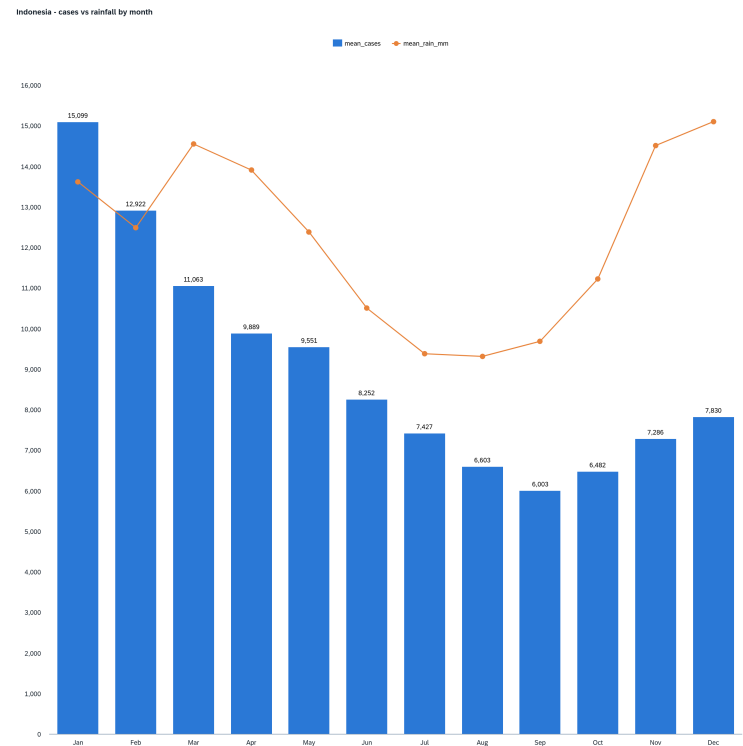
All charts built in SAP Analytics Cloud.

ANALYSIS 1 — THE SEASON

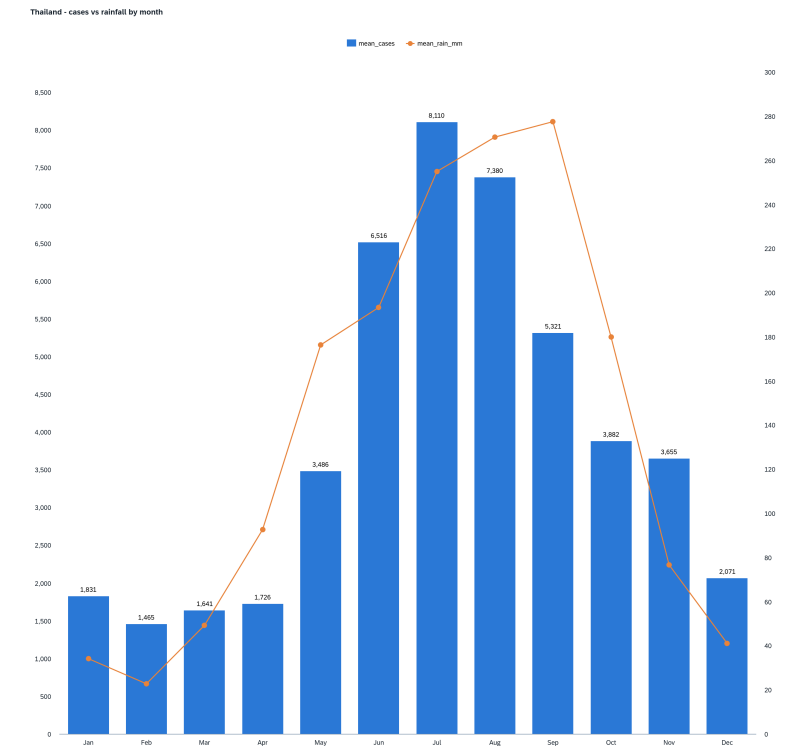
Dengue has a calendar — and it is not the same calendar everywhere

- **Indonesia:** cases peak **Jan–Apr** (Jan avg 15,099/month vs Sep 6,003 — $2.5\times$), right after the December rainfall peak
- **Thailand:** cases climb with the **onset** of the wet season and peak **Jun–Sep** (Jul avg 8,110 vs Feb 1,465 — $5.5\times$)
- Same disease, opposite halves of the year → a regional alarm must be **local in timing**

Indonesia — cases vs rainfall by month



Thailand — cases vs rainfall by month



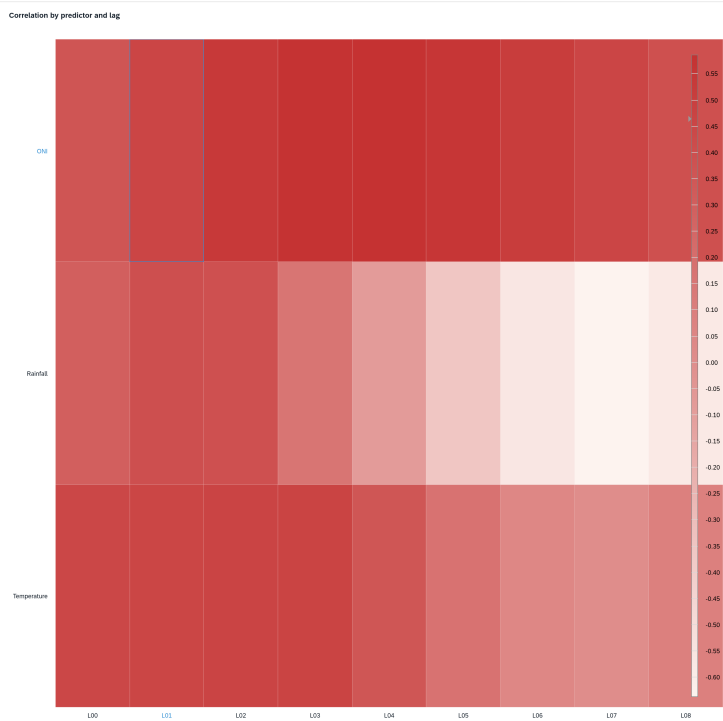
ANALYSIS 2 — THE LAG

The climate signal arrives 1–4 months before the cases

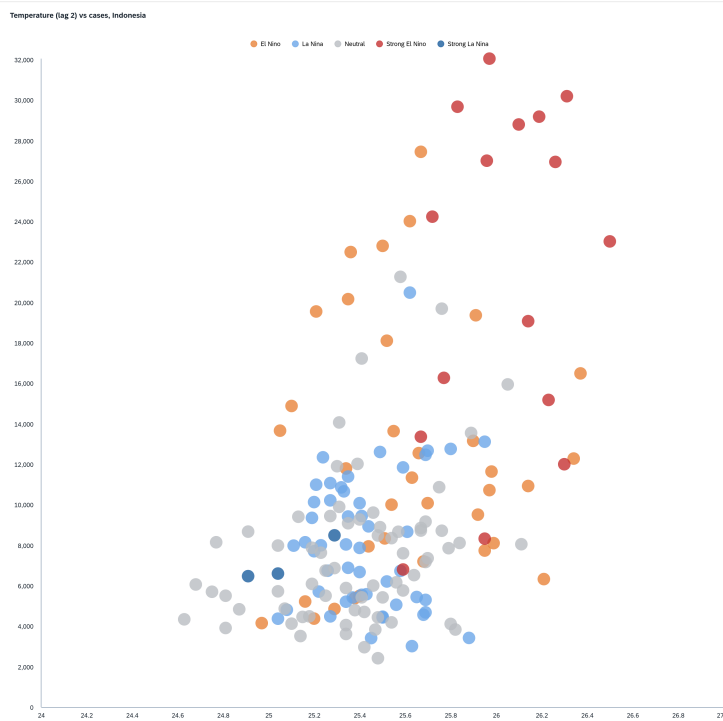
COUNTRY	BEST PREDICTOR	LAG	R
Indonesia	ONI (El Niño)	4 mo	+0.59
Indonesia	Temperature	3 mo	+0.47
Indonesia	Rainfall	1 mo	+0.40
Thailand	Temperature	3 mo	+0.57
Thailand	Rainfall	same mo	+0.48

The lag *is* the opportunity: it is the time a health office has to act.

Correlation by predictor and lag



Temperature (lag 2) vs cases, Indonesia



ANALYSIS 3 — THE ACE CARD

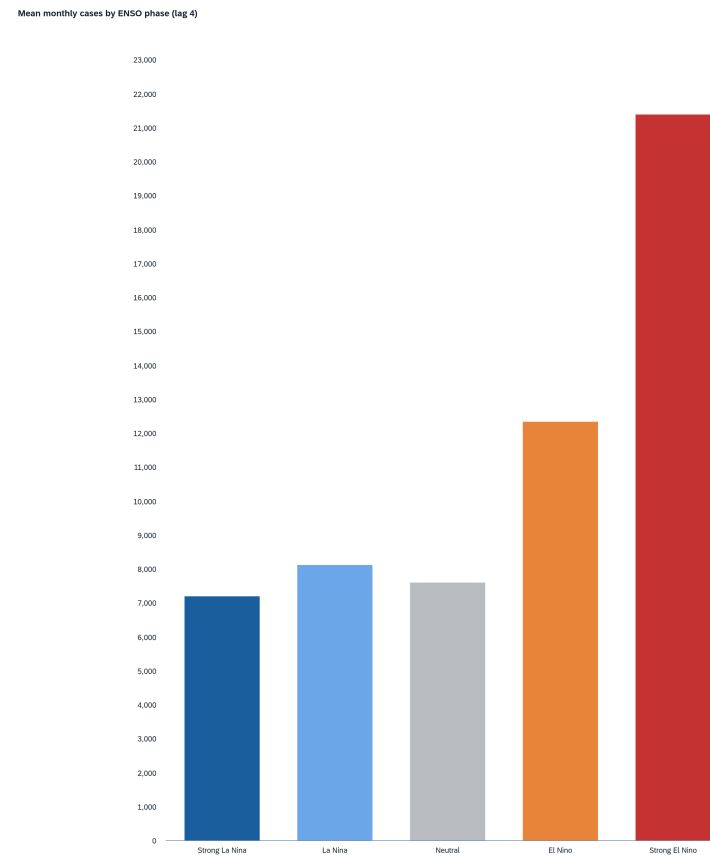
A strong El Niño nearly triples Indonesia's monthly dengue burden

2.8×

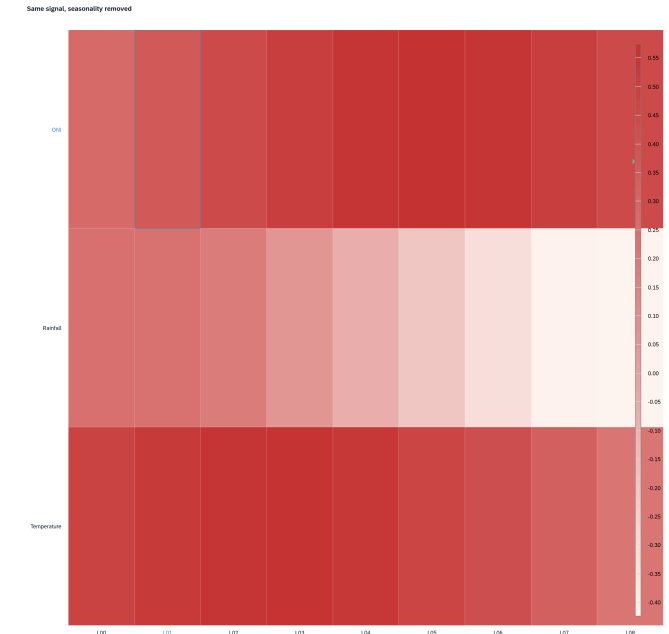
21,383 vs 7,597 cases per month, grouped by the ENSO phase four months earlier

- Grouped by the phase **four months earlier** — so this is a forecast, not hindsight
- **Not just seasonality**: after removing the seasonal cycle the ONI correlation holds at **$r = +0.56$**
- **Not an aggregation artefact**: cases are higher in El Niño months in **33 of 34 Indonesian provinces** (sign test $p = 2 \times 10^{-9}$) and 70 of 77 Thai provinces
- Rainfall does *not* survive the same test ($+0.40 \rightarrow +0.24$) — we report that honestly

Mean monthly cases by ENSO phase (lag 4)



Same signal, seasonality removed

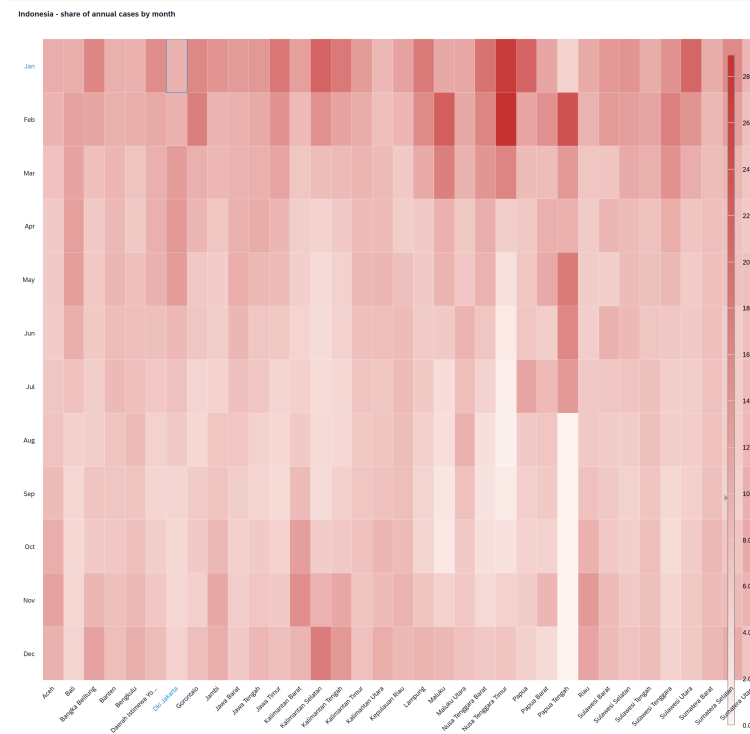


ANALYSIS 4 — WHERE AND WHEN

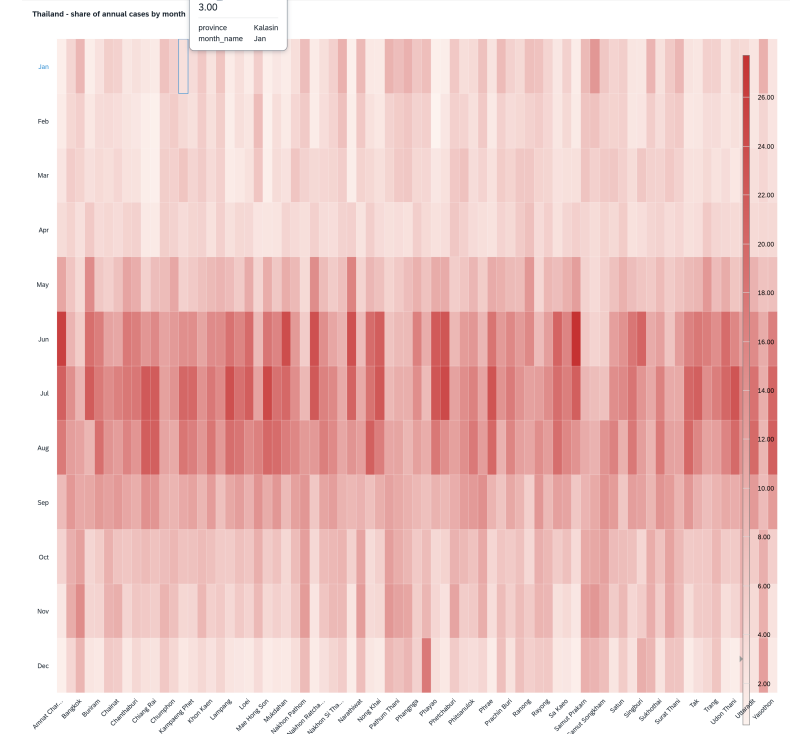
The alarm must be set province by province

- **Indonesia:** 20 of 35 provinces peak in **January**, 8 in February — but **Bali peaks in May** and North Kalimantan in December
- **Thailand:** 38 of 77 provinces peak in **July**, 16 in August
- A single national alert date would be wrong for a third of Indonesia

Indonesia — share of annual cases by month



Thailand — share of annual cases by month

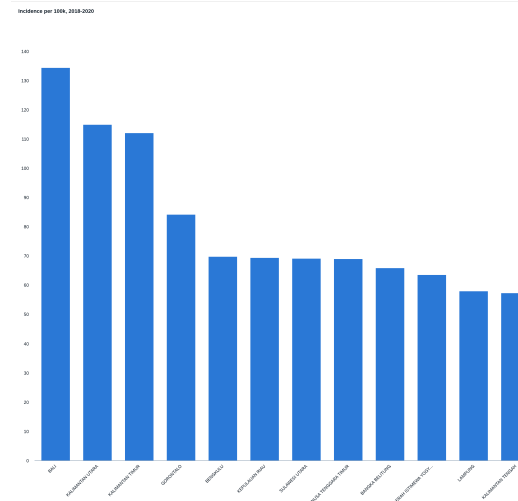


ANALYSIS 5 — WHO GETS HIT HARDEST

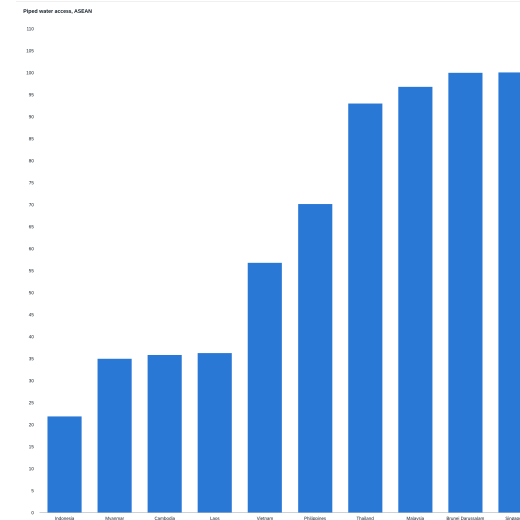
The same alarm, aimed at the most exposed places

- Highest incidence 2018–2020: **Bali 134**, North Kalimantan 115, East Kalimantan 112 per 100k — **14× the lowest province** (Papua, 9)
- Indonesia has the **lowest piped-water access in ASEAN: 21.8%** (JMP 2024) — households store water, and stored water breeds *Aedes*
- **12.9 million people in Bangkok** and 1.7 million in Jakarta live in the 1-in-100-year flood zone
- We use this to **target** the alarm, not to explain it (see p. 14)

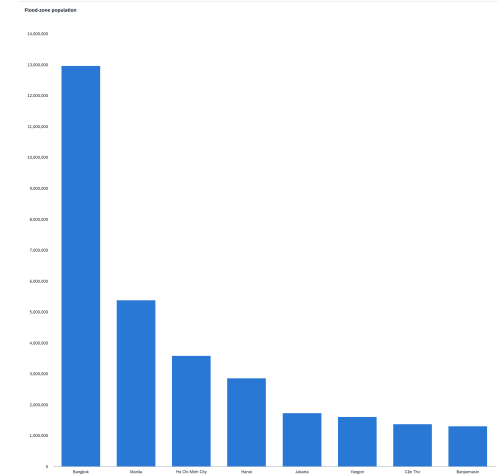
Incidence per 100k, 2018–2020



Piped water access, ASEAN



Flood-zone population

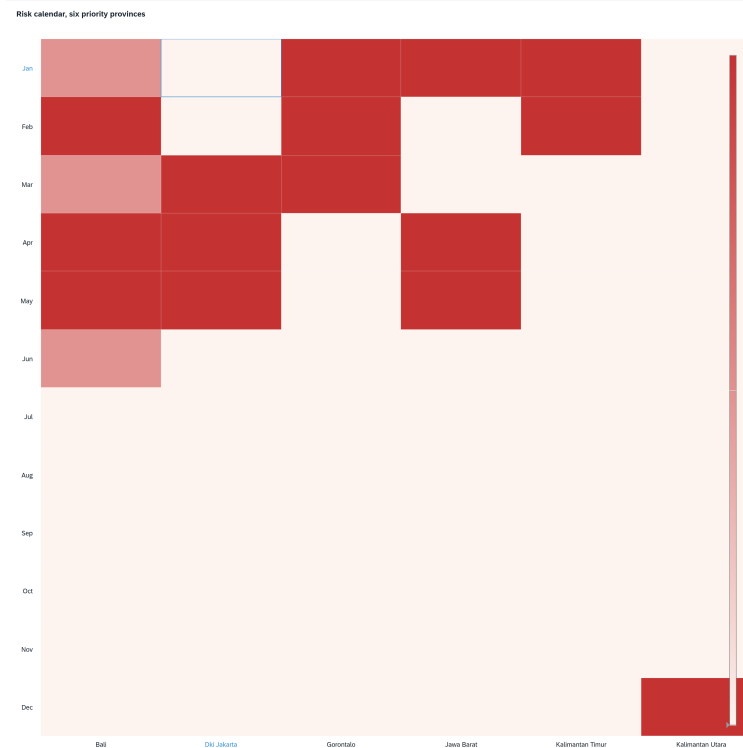


SYNTHESIS

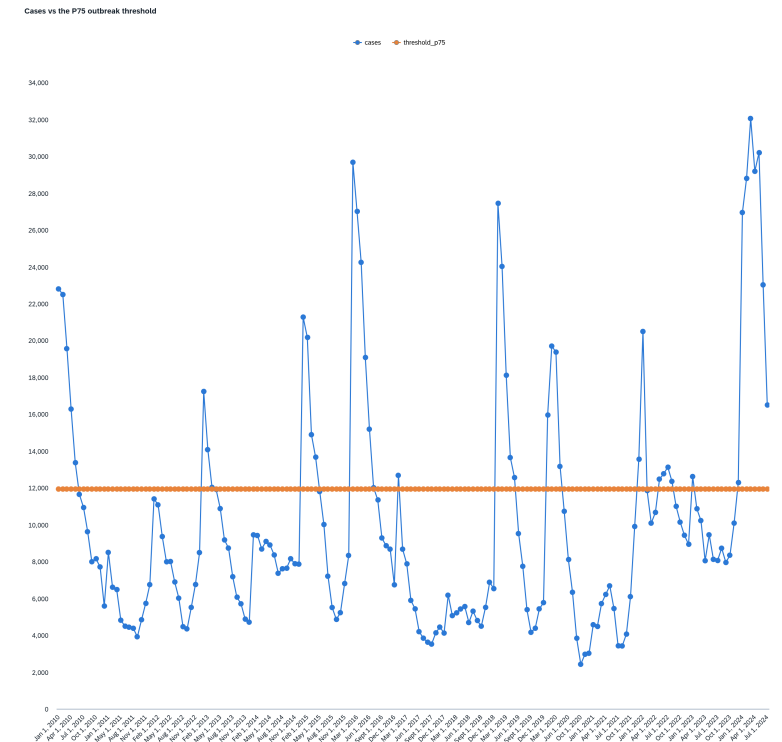
From signal to schedule: a risk calendar every health office can read

- Each province gets three states — **Peak** · **Watch** · **Normal** — from its own case history
- The El Niño index sets **how severe** the coming peak will be; the calendar sets **when** it arrives
- Real lead times, every El Niño episode since 2009: **0–3 months** on observed ONI — and longer on NOAA/BMKG *forecast* ONI

Risk calendar, six priority provinces



Cases vs the P75 outbreak threshold



THE SOLUTION

Two conditions. No model. No code.

“If the month is Jan–Apr AND ONI four months ago was $\geq 0.5 \rightarrow$ raise alert”

11.6%

of months fire the alert

100%

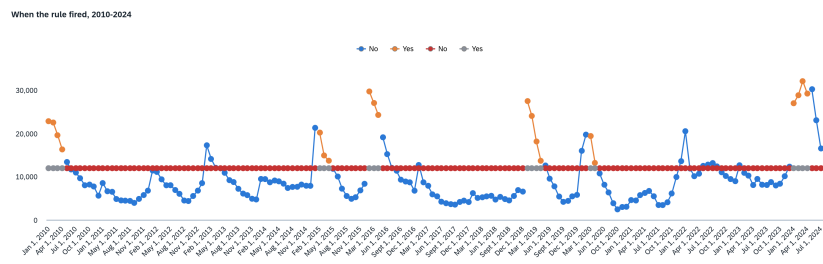
precision — every alert was a top-quartile dengue month

4×

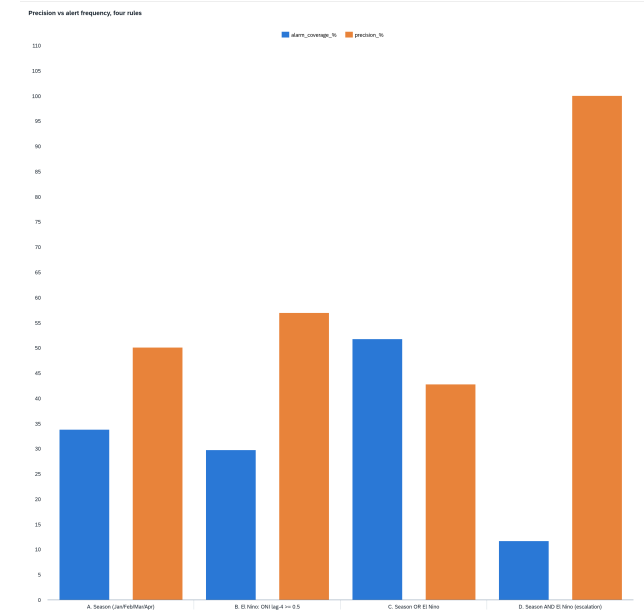
lift over the base rate

- Both inputs are public and free: NOAA publishes ONI monthly, the calendar comes from the office's own records
- Runs in a spreadsheet, a WhatsApp broadcast, or this SAC dashboard — competitors promise machine learning; we promise something a provincial health office can start next week

When the rule fired, 2010–2024



Precision vs alert frequency, four rules

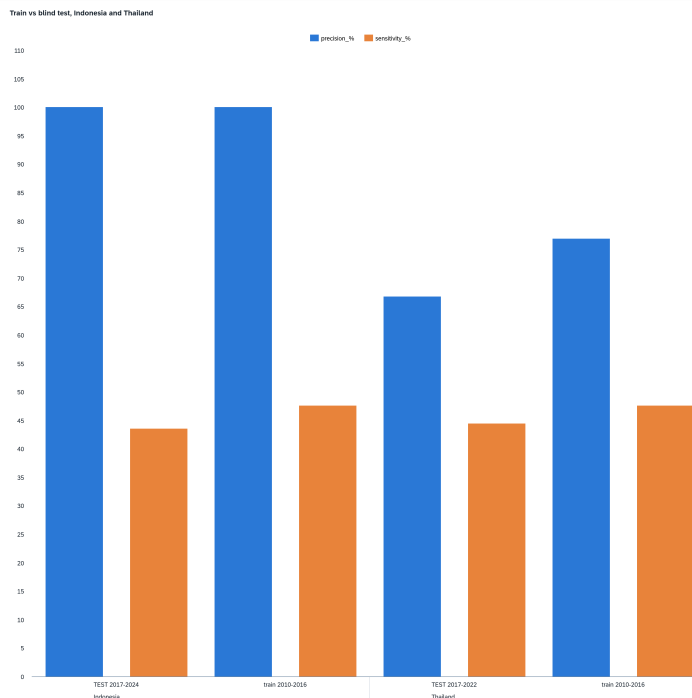


VIABILITY

We tried to break our own rule. It held.

- **Out-of-sample test:** rule built on 2010–2016, tested blind on 2017–2024 → **precision still 100%**, sensitivity 44%, lift 3.9
- Against the **WHO/PAHO endemic-channel** definition — which removes seasonality by construction — specificity **91%**, lift 2.5, and *all* of the value comes from the El Niño condition, not the season
- Published dengue EWS report PPV 43–86% (17-system review); we reach a comparable PPV **with no statistical model and monthly data**
- Replicates in Thailand with its own best predictor (temperature anomaly, lag 2): lift 2.7

Train vs blind test, Indonesia and Thailand



How we compare to published EWS

How we compare to published EWS

[system](#) | [1](#)

Measures		sensitivity_low	sensitivity_high	precision_low	precision_high
system	source				
17 published EWS (range)	Baharom et al. 2022	50	100	43	86
EWARS-csd, Colombia	Schlesinger et al. 2024	97	97	75	75
Rule D vs P75 (blind test)	this notebook	44	44	100	100
Rule D vs endemic channel	this notebook	29	29	30	30
WHO-TDR EWARS, Mexico	Cardenas et al. 2021	100	100	83	83

ADOPTION IN THREE PHASES

- 1 Bali, N. & E. Kalimantan pilot →
- 2 all 38 provinces →
- 3 ASEAN peers

Cost: open data, monthly refresh, existing staff.

IMPACT — INDONESIA ALONE

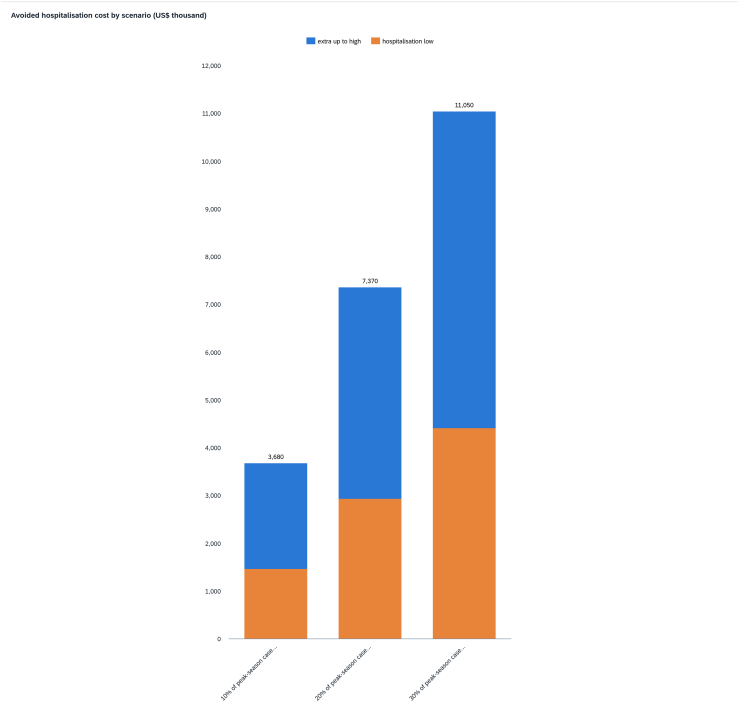
Acting 4–8 weeks earlier is worth US\$1.5–3.7 million a year

4,656

cases/year prevented in Indonesia, under the most conservative scenario

- Indonesia averages **104,753 cases/year** (complete years 2015–2023); **44% fall in Jan–Apr**
- Prevent just **10%** of peak-season cases (an assumption, not a finding) = 4,656 cases/year
- At Indonesia’s published hospitalisation cost of US\$316–791/case → **US\$1.5–3.7 M (Rp 24–59 billion)/year**
- For scale: dengue costs Indonesia **US\$381–681 M/year** (Nadjib 2019; Wilastonegoro 2020)
- Lead time is real: in the 2023–24 episode the alarm fired **Sep 2023**; cases crossed the peak threshold **Dec 2023**

Avoided hospitalisation cost by scenario



Lead time, every El Niño episode since 2009

Lead time, every El Nino episode since 2009

episode episode lead_time_months

episode	alarm	cases_cross	lead_time_months
2009/10	Dec 1, 2009 (2009)	Jan 1, 2010 (2010)	1
2014/16	Feb 1, 2015 (2015)	Feb 1, 2015 (2015)	0
2018/19	Jan 1, 2019 (2019)	Jan 1, 2019 (2019)	0
2019/20	Mar 1, 2020 (2020)	Mar 1, 2020 (2020)	0
2023/24	Sep 1, 2023 (2023)	Dec 1, 2023 (2023)	3

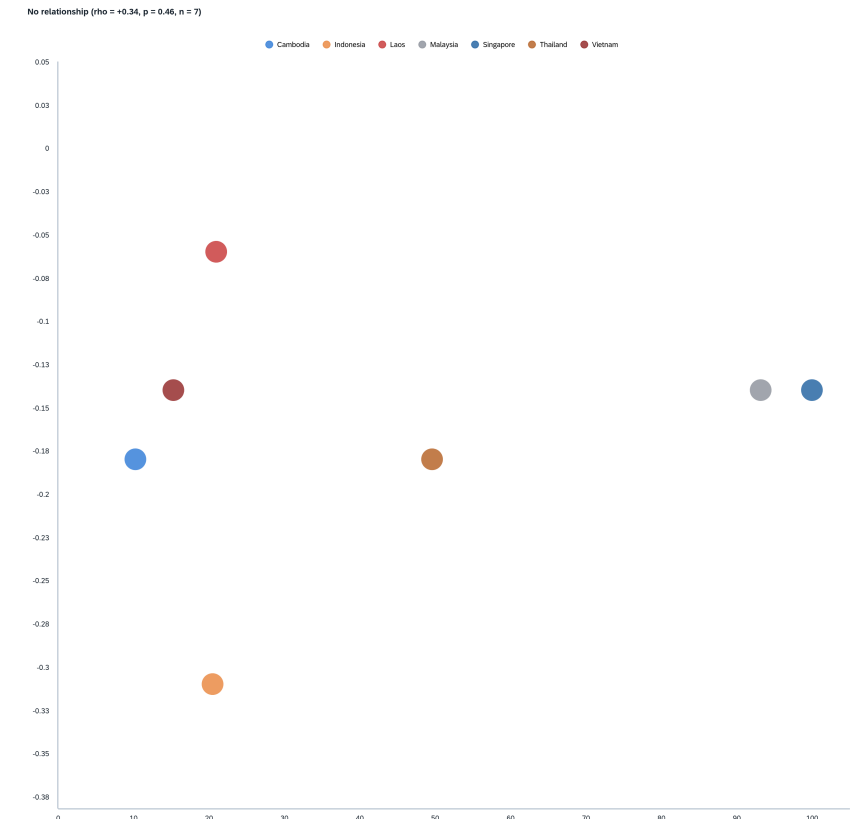
Sources: cost per case from peer-reviewed Indonesian studies; case counts from our own panel. The cheapest anchor (US\$90/episode) still gives US\$0.42 M — we did not pick the most flattering number.

LIMITS & DATA ETHICS

What we tested and threw away

- **We falsified our own innovation angle.** “Drought × water access” looked promising ($r = -0.31$) but collapsed after controlling for El Niño (partial $r = -0.15$) and showed no moderation by piped access ($\rho = +0.34$, $p = 0.46$, $n = 7$). **We dropped the claim.**
- **We do not claim a 2023–24 record** — our panel is incomplete for Vietnam and the Philippines after 2022.
- **Incidence only for 2018–2020**, the only years with official BPS denominators. Elsewhere we use raw counts and never rank regions by them.
- **Flood and disaster records look correlated with dengue ($r = +0.66$) — until you divide by population ($r = -0.14$).** We report the null.
- Correlation is not causation: this is an **early-warning** claim, with mechanism cited from the literature.

No relationship ($\rho = +0.34$, $p = 0.46$, $n = 7$)



THE PRODUCT

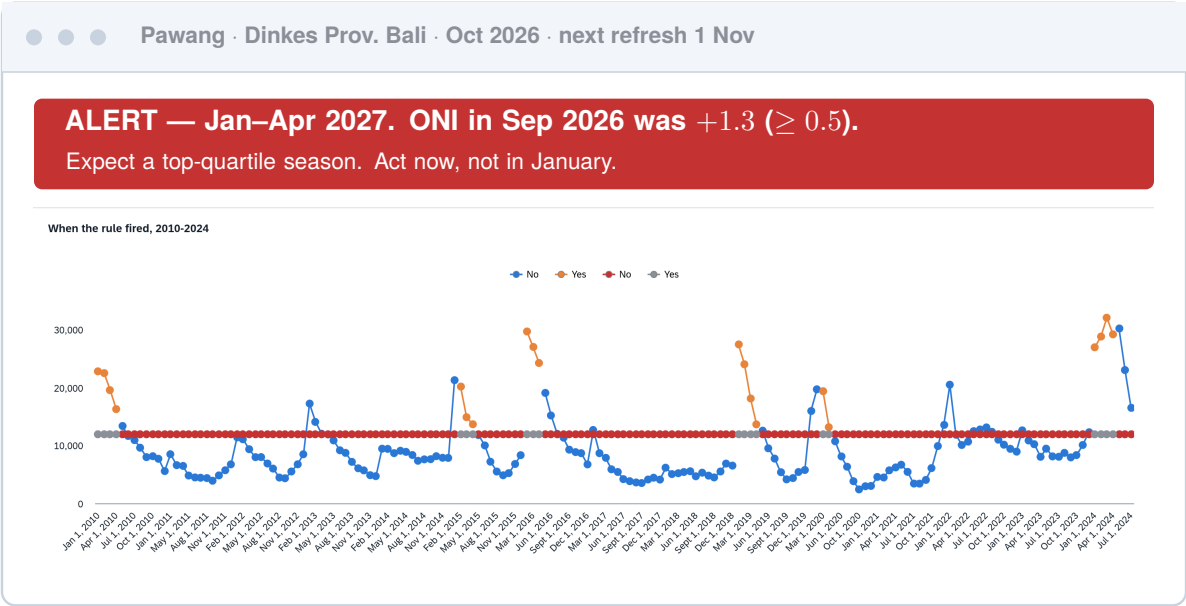
Pawang — the rule, delivered on the first of every month

Pawang

pawang — the Indonesian and Malay word for someone who claims to know the weather months ahead. This one shows its working: one SAC page a district officer opens on the 1st, plus an automatic WhatsApp advisory. It escalates; it never declares an emergency.

The digital layer does the fetching, not the deciding. SAC refreshes the NOAA ONI feed and the office’s own case calendar on a schedule, and SAC Smart Predict runs beside the rule as a second opinion. The alert itself stays the two-condition rule — auditable by hand, so nobody has to trust a model they cannot inspect.

What Bali opens on 1 October 2026



Who presses which button

WHO	ROLE
District health office gov't	Reads the advisory; moves PSN/3M, larviciding and stockpiles forward into the alerted window
Ministry of Health · BMKG gov't	Own the monthly refresh; BMKG’s forecast ONI stretches the notice past four months
Jumantik cadres NGO & community	Larval checks and household messaging concentrated in the alerted months instead of spread thin all year
Cloud & telco partners private	Host the SAC tenant and the broadcast — one analyst-hour a month, no data team

SCALES BY SWAPPING ONE INPUT

Thailand already validates with its own predictor (temperature anomaly, lag 2, lift 2.7). Any ASEAN member repeats the fit on its own records.

REFERENCES 1/2

Data sources

DATA

OpenDengue V1.3. doi:10.6084/m9.figshare.24259573.v4

World Bank Climate Change Knowledge Portal (ERA5 0.25° monthly temperature and precipitation).

NOAA Climate Prediction Center. Oceanic Niño Index (ONI v5).

WHO/UNICEF Joint Monitoring Programme for Water Supply, Sanitation and Hygiene, 2025 update.

BPS-Statistics Indonesia. Provincial population, WebAPI.

European Commission JRC. Global Human Settlement Layer (GHS-UCDB R2019A).

SIPSN, Ministry of Environment and Forestry, Indonesia.

BNPB. Indonesian Disaster Data and Information (DIBI).

METHOD REFERENCE

Hussain-Alkhateeb L, et al. (2017). *Early warning and response system (EWARS) for dengue outbreaks: operational guide*. WHO/TDR technical document — no DOI assigned; the peer-reviewed description is Hussain-Alkhateeb 2018 on the next page.

CLIMATE AND DENGUE

Tian Y, et al. (2025). Rising dengue risk with increasing El Niño–Southern Oscillation amplitude and teleconnections. *Nat Commun*. doi:10.1038/s41467-025-63655-0

Djaafara B, et al. (2026). Dengue transmission heterogeneity across Indonesia's archipelago: climate-driven spatiotemporal patterns and policy implications. *PLOS Negl Trop Dis*. doi:10.1371/journal.pntd.0014135

Mokhtar S, et al. (2024). Global risk of dengue outbreaks and the impact of El Niño events. *Environ Res*. doi:10.1016/j.envres.2024.119830

Sutriyawan A, et al. (2025). Impact of climate change on dengue incidence: a systematic review of evidence from Southeast Asia. *J Popul Soc Stud*. doi:10.25133/jpssv342026.028

Lowe R, et al. (2021). Combined effects of hydrometeorological hazards and urbanisation on dengue risk in Brazil. *Lancet Planet Health*. doi:10.1016/S2542-5196(20)30292-8

Gibb R, et al. (2023). Interactions between climate change, urban infrastructure and mobility are driving dengue emergence in Vietnam. *Nat Commun*. doi:10.1038/s41467-023-43954-0

Wang Y, et al. (2022). Impact of extreme weather on dengue fever infection in four Asian countries. *Environ Int*. doi:10.1016/j.envint.2022.107518

Choi Y, et al. (2016). Effects of weather factors on dengue fever incidence and implications for interventions in Cambodia. *BMC Public Health*. doi:10.1186/s12889-016-2923-2

Polrob P, et al. (2025). Nonlinear and lagged effects of climate variability on dengue incidence in Bangkok. *BMC Public Health*. doi:10.1186/s12889-025-25420-2

REFERENCES 2/2

Early-warning systems, cost, and validation

EARLY-WARNING SYSTEMS

Hussain-Alkhateeb L, et al. (2018). Early warning and response system (EWARS) for dengue outbreaks: recent advancements towards widespread applications in critical settings. *PLOS ONE*. doi:10.1371/journal.pone.0196811

Baharom M, et al. (2022). Dengue early warning system as outbreak prediction tool: a systematic review. *Risk Manag Healthc Policy*. doi:10.2147/RMHP.S361106

Benitez-Valladares D, et al. (2021). Validation of the early warning and response system (EWARS) for dengue outbreaks: evidence from the national vector control program in Mexico. *PLOS Negl Trop Dis*. doi:10.1371/journal.pntd.0009261

Cárdenas R, et al. (2022). The early warning and response system (EWARS-TDR) for dengue outbreaks: can it also be applied to chikungunya and Zika outbreak warning? *BMC Infect Dis*. doi:10.1186/s12879-022-07197-6

Schlesinger P, et al. (2024). Enabling countries to manage outbreaks: statistical, operational, and contextual analysis of the EWARS-csd for dengue outbreaks. *Front Public Health*. doi:10.3389/fpubh.2024.1323618

Quan TM, et al. (2026). Climatic determinants and simple thresholds for dengue early warning in Vietnam: a One Health perspective. *Sci One Health*. doi:10.1016/j.soh.2026.100159

BURDEN, COST, AND VALIDATION

Nadjib M, et al. (2019). Economic burden of dengue in Indonesia. *PLOS Negl Trop Dis*. doi:10.1371/journal.pntd.0007038

Wilastonegoro NN, et al. (2020). Cost of dengue illness in Indonesia across hospital, ambulatory, and not medically attended settings. *Am J Trop Med Hyg*. doi:10.4269/ajtmh.19-0855

Shepard DS, et al. (2013). Economic and disease burden of dengue in Southeast Asia. *PLOS Negl Trop Dis*. doi:10.1371/journal.pntd.0002055

Shepard DS, et al. (2016). The global economic burden of dengue: a systematic analysis. *Lancet Infect Dis*. doi:10.1016/S1473-3099(16)00146-8

Sasmono RT, et al. (2026). Understanding the epidemiological trends and economic impact of dengue in Indonesia. *IJID Regions*. doi:10.1016/j.ijregi.2026.100860

Childs ML, et al. (2025). Climate warming is expanding dengue burden in the Americas and Asia. *PNAS*. doi:10.1073/pnas.2512350122